1 L2/L3 HH Warm-Start PAOP+HVA Results (Human-Readable PDF Report)

1.1 1. Parameter Manifest

- Model family/name: `Hubbard` (HH specialization).
- Ansatz types covered: `hh_hva_tw`, `hh_hva_ptw`, ADAPT(`paop_lf_std`), ADAPT trend pools (`uccsd+paop`, `uccsd+paop+hva`).
- Drive enabled: both `false` (static sections) and `true` (driven section from `drive_from_fix1_warm_start_B_full.json`).
- Core physical parameters: L2 static uses `t=1.0, U=2.0, dv=0.0`; L3 runs use `t=1.0, U=4.0, dv=0.0`.
- HH-defining parameters: `omega0=1.0`, `g_ep={1.0 (L2), 0.5 (L3)}`, `n_ph_max=1`, boson encoding `binary`, ordering `blocked`, boundary `open`.
- Reproducibility sector references: L2 `(N_up,N_dn)=(1,1)`; L3 `(N_up,N_dn)=(2,1)`.
- Driven-run timing/drive: `t_final=15.0`, `num_times=201`, `trotter_steps=192`, `exact_steps=384`, `A=0.5`, `omega=2.0`, `phi=0.0`, `t0=0.0`, `tbar=3.0`, pattern `staggered`.

1.1.1 1.1 What "Accessibility Rerun" Means (Exhaustive Definition)

In this report, "Accessibility rerun" does **not** mean a UI/accessibility feature and does **not** mean driven Trotter dynamics. It means a second-pass static ADAPT optimization run where we keep a fixed practical compute budget and re-run the warm-start branch under those fixed limits to measure what quality is realistically reachable.

Concretely, the accessibility runs come from `ACC_JSON = artifacts/useful/L3/l3_hh_accessibility_fixes_under8pct.jsor` Each rung (for example, `Acc B` and `Acc C`) specifies explicit ADAPT limits such as `adapt_max_depth`, `adapt_maxiter`, `eps_grad`, and `wallclock_cap_s=1200`. The purpose is to compare outcomes under controlled run limits, not to change physics definitions.

The non-accessibility counterpart in this PDF is `B export` from `B_EXPORT_JSON = artifacts/useful/L3/warmstart_states/fi` which is the direct exported rebuild artifact for warm-start branch B. That is why `B export` and `Acc B` can be very close in `DeltaE` but still differ in recorded depth/parameter count (`42` vs `43`) and runtime fields: they are separate executions with separate bookkeeping.

For this reason, throughout this document: - `Accessibility rerun` = fixed-budget static ADAPT rerun (`Acc B`, `Acc C`). - `Driven dynamics` = time-evolution branch analysis from `drive_from_fix1_warm_start_B_full.json`.

1.2 2. Executive Summary

- L2 strong VQE reaches `DeltaE=6.029e-07` (near exact).
- L3 warm VQE alone remains at `DeltaE=1.969e-02`.
- Warm-start + ADAPT improves L3 to `DeltaE=4.393e-03` (Accessibility C).
- Trend runs reach `DeltaE~2.622e-04` (best shown point).
- Drive branch diagnostics show PAOP branch fidelity near `0.995893` vs HVA near `0.827750` against the same filtered exact manifold.

1.3 3. Math and Metrics (Compact)

$$H = H_t + H_U + H_{ph} + H_{e-ph}, \quad H(t) = H + H_{drive}(t).$$

$$E_{exact,sector} = \min_{\psi \in \mathcal{H}_{(N_\uparrow, N_\downarrow)}} \langle \psi | H | \psi \rangle.$$

$$\Delta E = |E_{best} - E_{exact,sector}|, \quad \varepsilon_{rel} = \Delta E / |E_{exact,sector}|.$$

$$|\psi_d\rangle = e^{-i\theta_d G_d} \cdots e^{-i\theta_1 G_1} |\psi_0\rangle, \quad g_m = i\langle \psi_d | [H, G_m] | \psi_d \rangle.$$

Cost metrics used in this report:

$$\kappa_{eval} = nfev/P, \quad \tau_{eval} = runtime_s/nfev.$$

1.4 4. Run Labels and Measurement Dictionary

1.4.1 4.1 Run Label Glossary (What each row name means)

Label in tables/figures	Provenance ID	Definition used in this report
`L2 VQE`	`L2_JSON`	Hardcoded HH VQE on L=2 (`hh_hva_tw`, `L-BFGS-B`).
`L3 warm VQE`	`L3_WARM_JSON`	Warm-start seed VQE stage on L=3 (`hh_hva_ptw`, `COBYLA`).
`B export`	`B_EXPORT_JSON`	Exported fix1 warm-start branch B state (`depth=42`, `pool=paop_lf_std`).
`Acc B`	`ACC_JSON[rung=B]`	Accessibility rerun for branch B (separate execution from `B export`, reached `depth=43`).
`Acc C`	`ACC_JSON[rung=C]`	Accessibility rerun for branch C (different rung settings).
`A_medium`	`TREND_JSON[A_medium]`	Trend pool A (`uccsd+paop`), medium budget (`depth=20`).
`A_heavy`	`TREND_JSON[A_heavy]`	Trend pool A (`uccsd+paop`), heavy budget (`depth=36`).
`B_medium`	`TREND_JSON[B_medium]`	Trend pool B (`uccsd+paop+hva`), medium budget (`depth=20`).
`B_heavy`	`TREND_JSON[B_heavy]`	Trend pool B (`uccsd+paop+hva`), heavy budget (`depth=36`).

Provenance key: - `L2_JSON` = `artifacts/useful/L2/H_L2_hh_termwise_regular_lbfgs_t1.0_U2.0_g1_nph1.json` - `L3_WARM_JSON` = `artifacts/json/hh_L3_hh_hva_ptw_heavy.json` - `B_EXPORT_JSON` = `artifacts/useful/L3/warmstart_st` - `ACC_JSON` = `artifacts/useful/L3/l3_hh_accessibility_fixes_under8pct.json` - `TREND_JSON` = `artifacts/useful/L3/l3_uccsd_paop_hva_trend_full_20260302T000521.json`

Clarification on naming: - In this report, `Acc B` means "Accessibility rung B" (same concept you referred to as "BACC"). - `B export` and `Acc B` are related but not identical runs; that is why depth can appear as `42` vs `43`.

1.4.2 4.2 Measurement Dictionary (What each metric is)

Metric label	Formula / source field(s)	Meaning	Units
`E_best`	`vqe.energy` or `result.E_best`	Best energy for that run definition.	energy
`E_exact`	`ground_state.exact_energy_filtered` or `exact.E_exact_sector`	Selected exact oracle used in `DeltaE`.	energy

Metric label	Formula / source field(s)	Meaning	Units
`DeltaE`	`|E_best - E_exact|`	Absolute static energy error.	energy
`eps_rel`	`DeltaE / |E_exact|`	Relative static energy error magnitude.	unitless
`Params P`	`vqe.num_parameters` or `result.num_parameters`	Number of optimized parameters in the final model.	count
`nfev`	`vqe.nfev` or `result.nfev_total`	Objective evaluations consumed in that run.	count
`Depth`	VQE proxy=`reps`; ADAPT depth field (`ansatz_depth` or `adapt_depth_reached`)	Ansatz growth indicator used for comparison.	count
`Runtime (s)`	`runtime_s` (or nearest runtime field in artifact)	Wall-clock runtime for that run block.	seconds
`F_paop` / `F_hva`	`fidelity_paop_trotter`, `fidelity_hva_trotter`	Branch fidelity versus filtered exact GS manifold projector.	unitless
`dE_total_paop` / `dE_total_hva`	`|E_total_exact - E_total_trotter|` (per branch)	Branch-wise total-energy Trotter error at each time.	energy
`dD_paop` / `dD_hva`	`|D_exact - D_trotter|` (per branch)	Branch-wise doublon observable error at each time.	unitless
`kappa_eval`	`nfev / P`	Evaluation pressure per free parameter.	eval/param
`tau_eval`	`runtime_s / nfev`	Average wall-clock cost per evaluation.	s/eval

1.5 5. Readable Static Scoreboard

Run	DeltaE	Depth	Params P	nfev	E_best	E_exact	eps_rel	Optimizer
L2 VQE	6.029e-07	6	108	15042	-0.389552500	-0.389553103	1.548e-06	L-BFGS-B
L3 warm VQE	1.969e-02	3	39	4000	0.264627255	0.244940700	8.037e-02	COBYLA
B export	6.508e-03	42	42	n/a	0.251448234	0.244940700	2.657e-02	ADAPT
Acc B	6.508e-03	43	43	8063	0.251448234	0.244940700	2.657e-02	ADAPT
Acc C	4.393e-03	38	38	6227	0.249334000	0.244940700	1.794e-02	ADAPT
A_medium	2.622e-04	20	20	12640	0.245202940	0.244940700	1.071e-03	ADAPT
A_heavy	2.629e-04	36	36	26833	0.245203648	0.244940700	1.074e-03	ADAPT
B_medium	2.622e-04	20	20	12640	0.245202940	0.244940700	1.071e-03	ADAPT
B_heavy	2.629e-04	36	36	26833	0.245203648	0.244940700	1.074e-03	ADAPT

Note: this table intentionally rounds values for readability; all source values are retained in JSON artifacts.

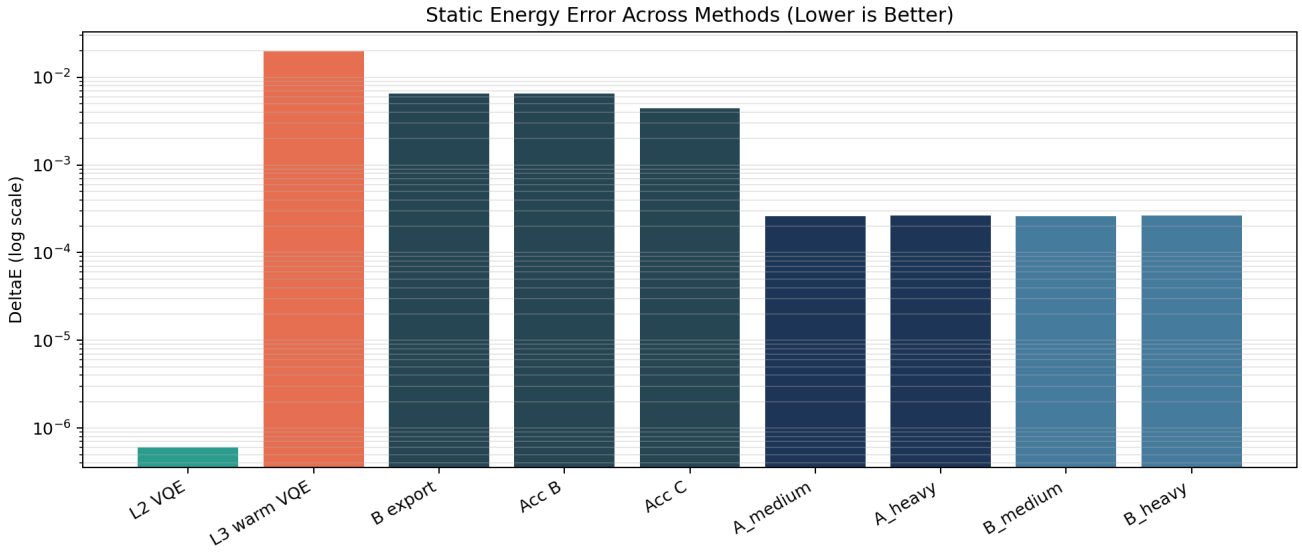

Figure 1: Static error comparison

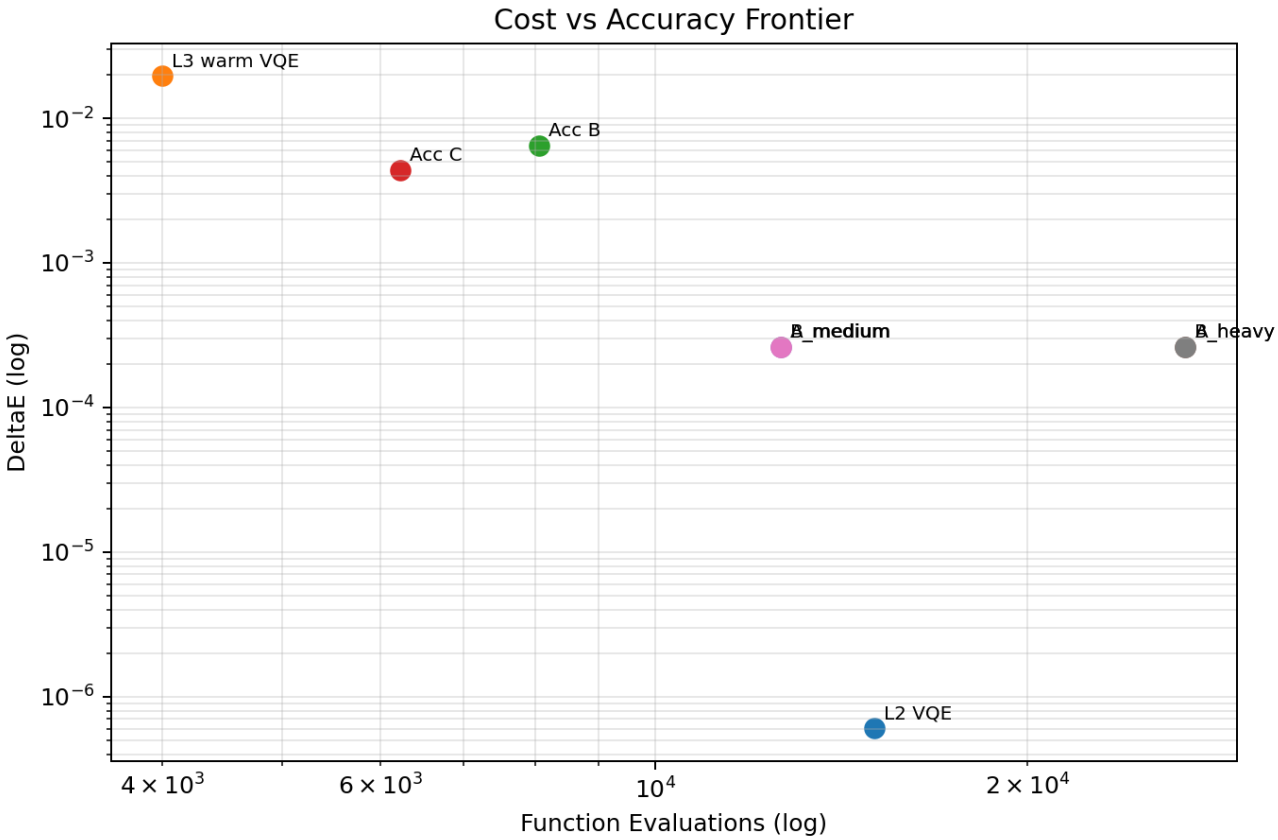

Figure 2: Cost vs accuracy

1.6 6. Accessibility and Warm-Start Convergence

Run	DeltaE_best	Depth	Params P	nfev	DeltaE_warm	Improvement factor DeltaE_warm/DeltaE_best	E_warm	E_best	Runtime (s)
B export	6.508e-03	42	42	n/a	1.969e-02	3.025	0.264627255	0.251448234	1201.4
Acc B	6.508e-03	43	43	8063	1.969e-02	3.025	0.264627255	0.251448234	1261.9
Acc C	4.393e-03	38	38	6227	1.234e-02	2.809	0.257282637	0.249334000	1208.3

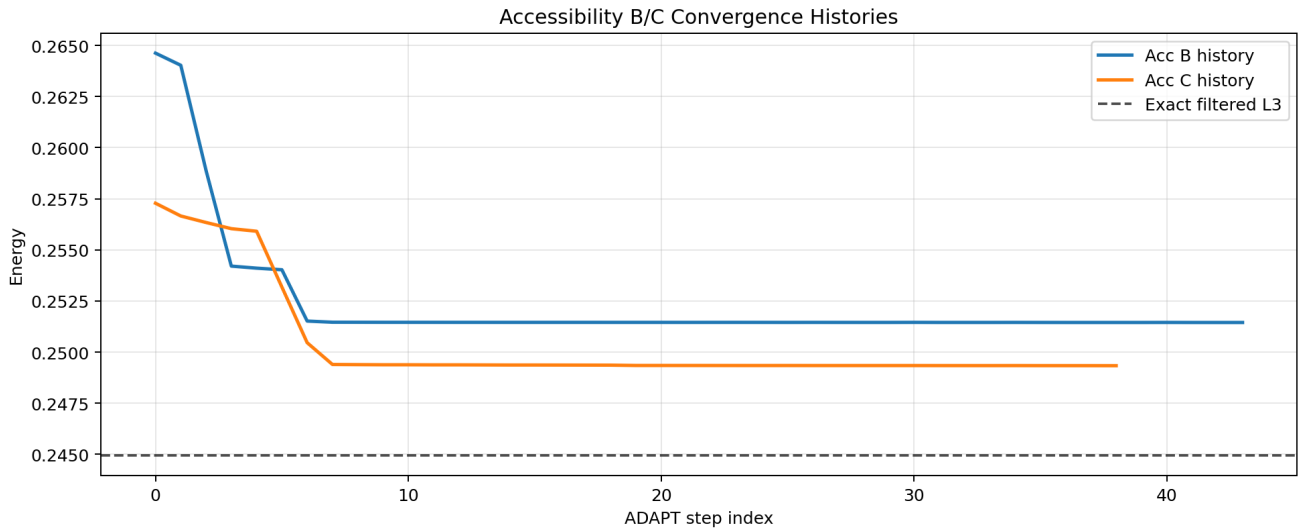

Figure 3: Accessibility convergence histories

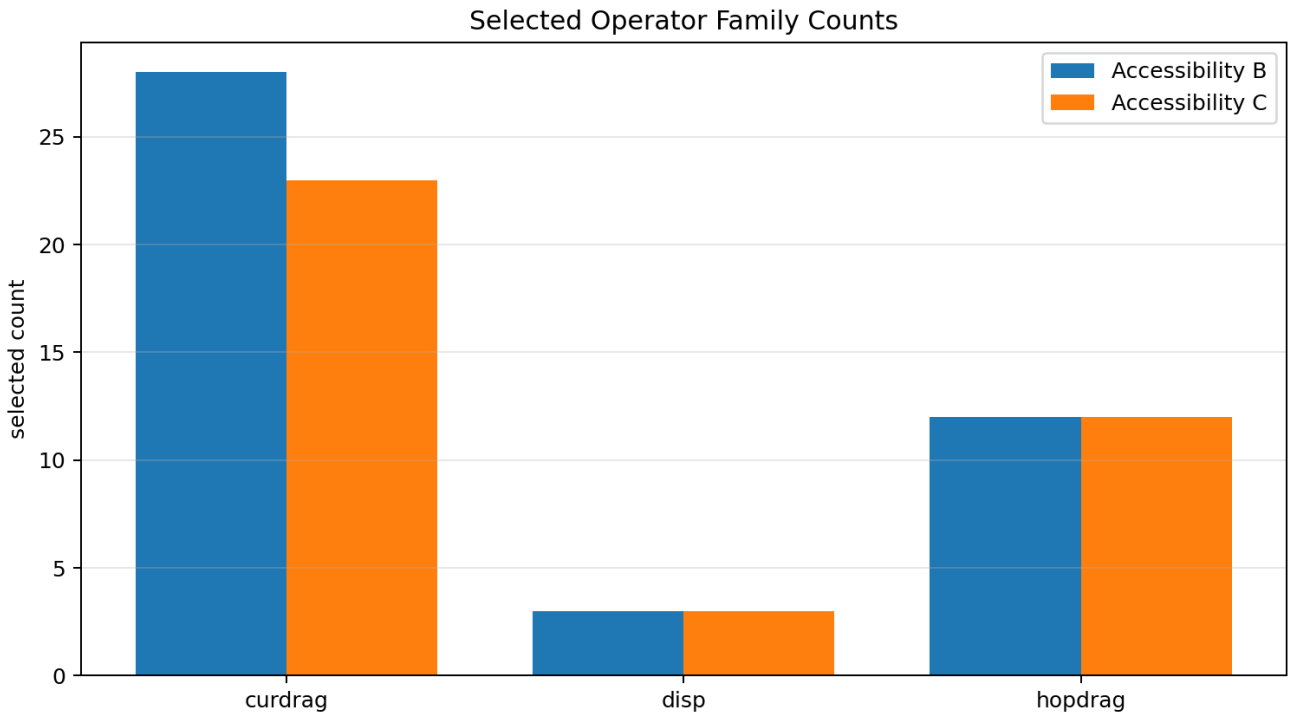

Figure 4: Selected operator family counts

1.7 7. Trend Experiment (A/B pools)

- Raw pool sizes: `UCCSD=8`, `PAOP=7`, `HVA=13`.
- Pool A (`uccsd+paop`): `dedup_total=15`, `overlap_count=0`.
- Pool B (`uccsd+paop+hva`): `dedup_total=28`, `overlap_count=0`.

Trend run	DeltaE	Depth	Params P	nfev	runtime (s)	final max
A_medium	2.622e-04	20	20	12640	624.0	4.564e-03
A_heavy	2.629e-04	36	36	26833	1722.5	7.722e-03
B_medium	2.622e-04	20	20	12640	720.7	4.564e-03
B_heavy	2.629e-04	36	36	26833	2379.3	7.722e-03

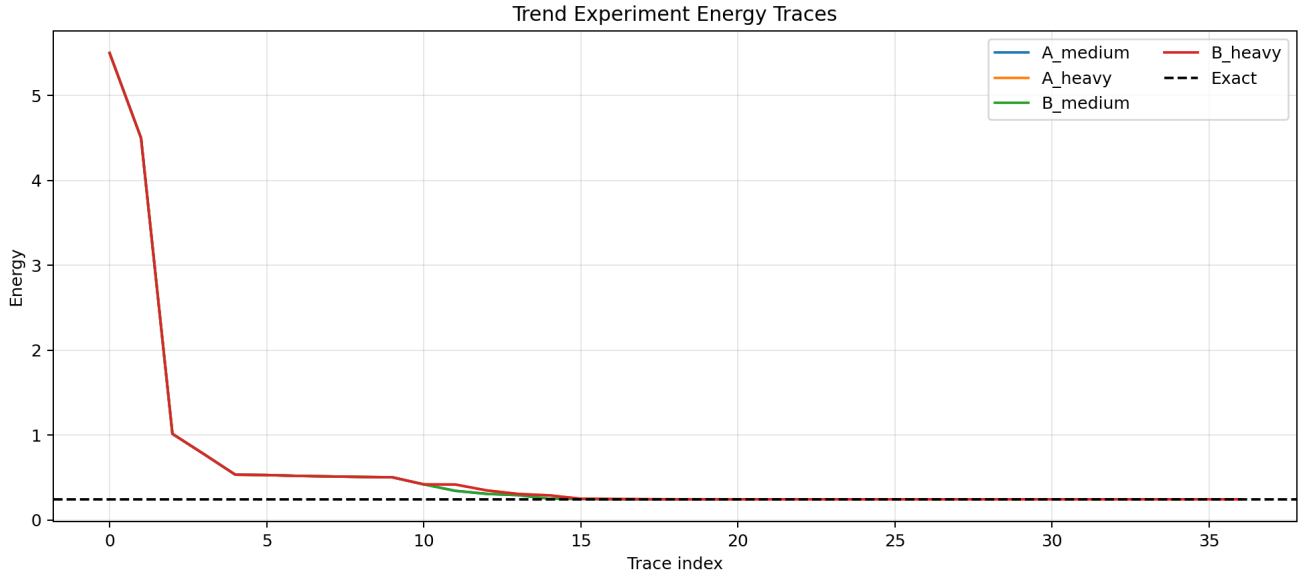

Figure 5: Trend energy traces

Section role boundary: Section 7 is static ADAPT trend optimization; Section 8 is driven branch dynamics.

1.8 8. Drive Dynamics (Branch-Aware)

Branch order: `exact_gs_filtered`, `exact_paop`, `trotter_paop`, `exact_hva`, `trotter_hva`.

1.8.1 8.0 Dynamics Reference Trajectory and How It Is Generated

For dynamics, the reference is the time-dependent exact trajectory

$$E_{ref}(t) \equiv E_{exact_gs_filtered}(t) = textttenergy_total_exact(t),$$

taken from the `exact_gs_filtered` branch in `drive_from_fix1_warm_start_B_full.json`.

This is not a single static scalar; it is a full time series used as the baseline for branch comparisons. In this run: - `E_ref(t=0) = 0.24494070012791422` - `E_ref(t=t_final=15) = 0.26953974487978255`

Repo options/fields used to generate this dynamics dataset (same file, `settings.drive` + top-level settings): - drive enabled: `true` - waveform: `A=0.5`, `omega=2.0`, `phi=0.0`, `t0=0.0`, `tbar=3.0`, pattern=`staggered` - Trotter setup: `suzuki_order=2`, `trotter_steps=192` - exact reference setup: `reference_steps_multiplier=2`, `reference_steps=384` - exact reference method: `exponential_midpoint_magnus2_order2` - backend: `scipy_sparse_expm_multiply`

1.8.2 8.1 Exact-GS vs PAOP/Trotter-PAOP (Primary Dynamics Comparison)

Naming map used in this section (`drive_from_fix1_warm_start_B_full.json`): - `exact_gs_filtered`: exact/reference propagation from filtered-sector exact ground-state initialization. - `exact_paop`: exact propagation from imported PAOP ADAPT state (`source=adapt_json`, `pool=paop_lf_std`, `depth=42`). - `trotter_paop`: Trotter propagation from the same PAOP initialization. - `exact_hva`: exact propagation from regular hardcoded VQE (`source=regular_vqe`, `ansatz=hh_hva_ptw`). - `trotter_hva`: Trotter propagation from the same HVA initialization.

Primary PAOP triad (total-energy distances):

Metric pair	t=0	mean over time	max over time	t=t_final
`|E_gs - E_exact_paop|`	6.508e-03	6.941e-03	7.938e-03	6.923e-03
`|E_gs - E_trotter_paop|`	6.508e-03	6.936e-03	7.910e-03	6.789e-03
`|E_exact_paop - E_trotter_paop|`	0.000e+00	1.380e-04	1.014e-03	1.340e-04

Companion HVA comparison (same metric form):

Metric pair	t=0	mean over time	max over time	t=t_final
`|E_gs - E_exact_hva|`	1.777e-01	1.859e-01	1.905e-01	1.876e-01
`|E_gs - E_trotter_hva|`	1.777e-01	1.859e-01	1.904e-01	1.868e-01
`|E_exact_hva - E_trotter_hva|`	0.000e+00	4.360e-04	2.766e-03	7.359e-04

Fidelity summary (vs filtered exact GS manifold projector):

Branch	mean	min	max	final
`paop`	0.995893	0.995732	0.996043	0.995881
`hva`	0.827750	0.827244	0.828695	0.828071

Driven overlays requested in this revision: - Plot A combines absolute energies and energy-error trajectories in one figure. - Plot B combines absolute energies, fidelities, and observables in one figure.

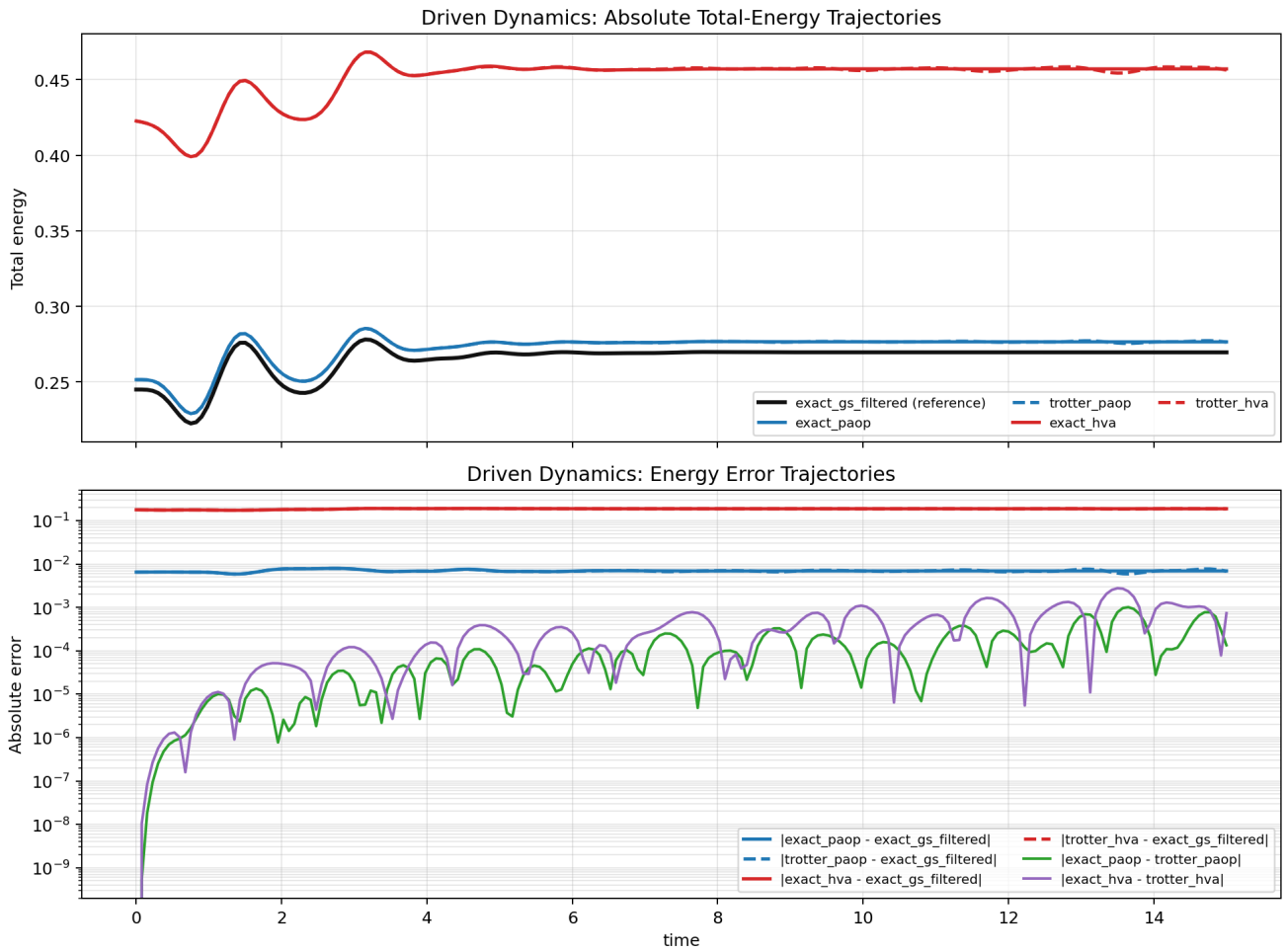

Figure 7: Drive absolute energies + energy errors

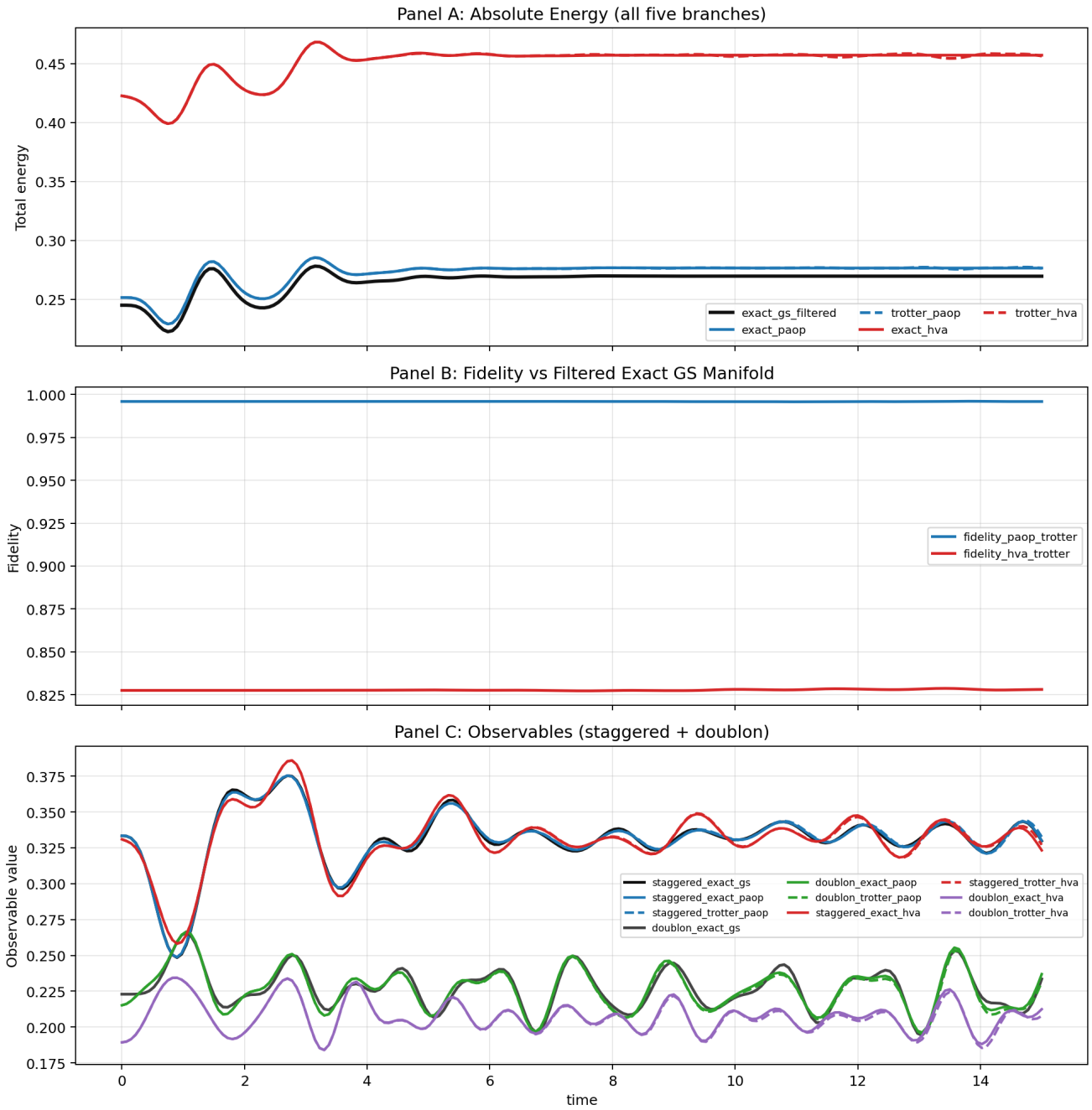

Figure 8: Drive energies + fidelities + observables

1.8.3 8.2 Additional Branch Aggregates

Metric	PAOP branch	HVA branch
Fidelity mean	0.995892975	0.827750111
Fidelity final	0.995881343	0.828070675
max	Delta E_total	
mean	Delta E_total	
max	Delta doublon	

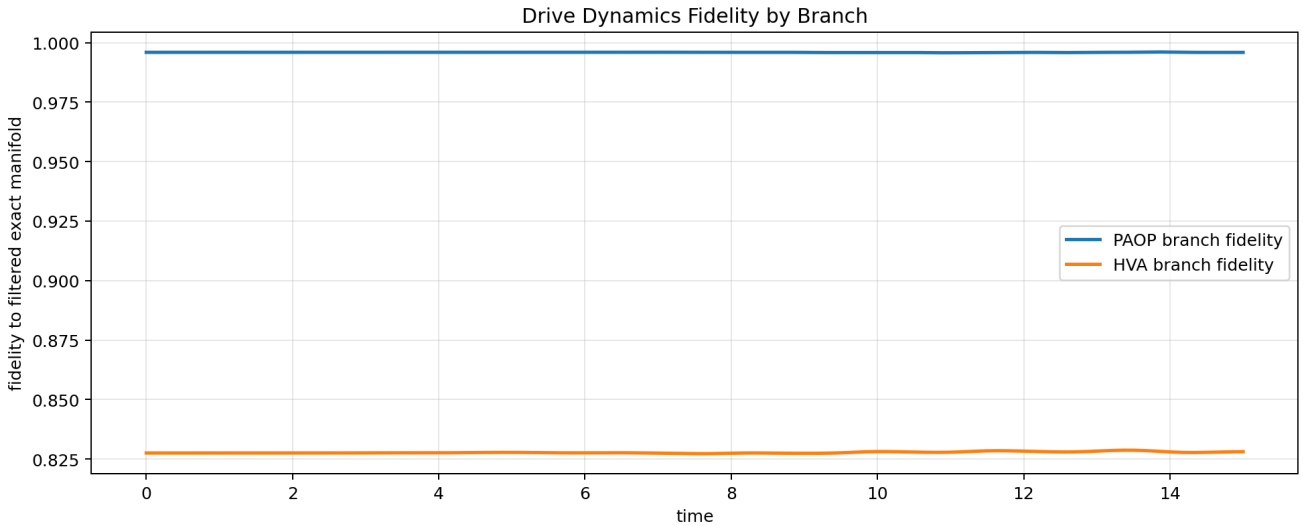

Figure 9: Drive fidelity

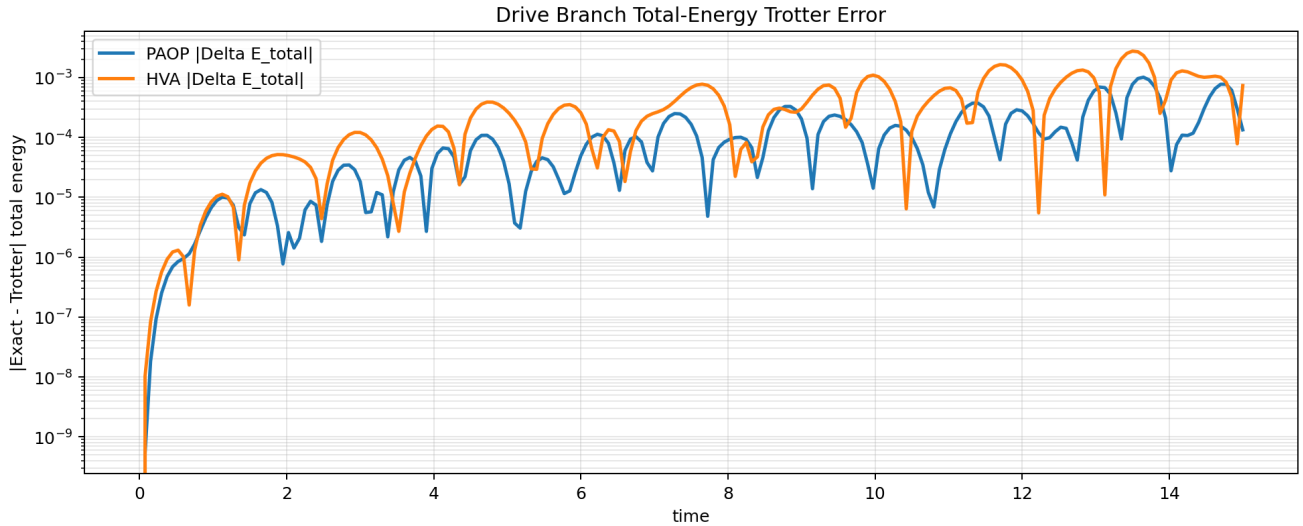

Figure 10: Drive total-energy error

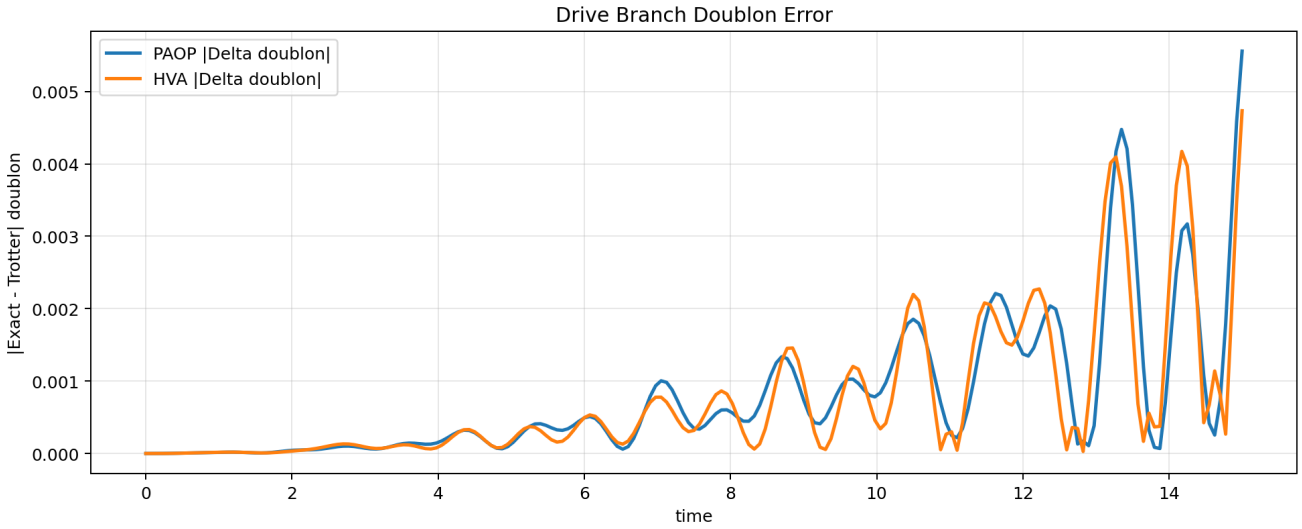

Figure 11: Drive doublon error

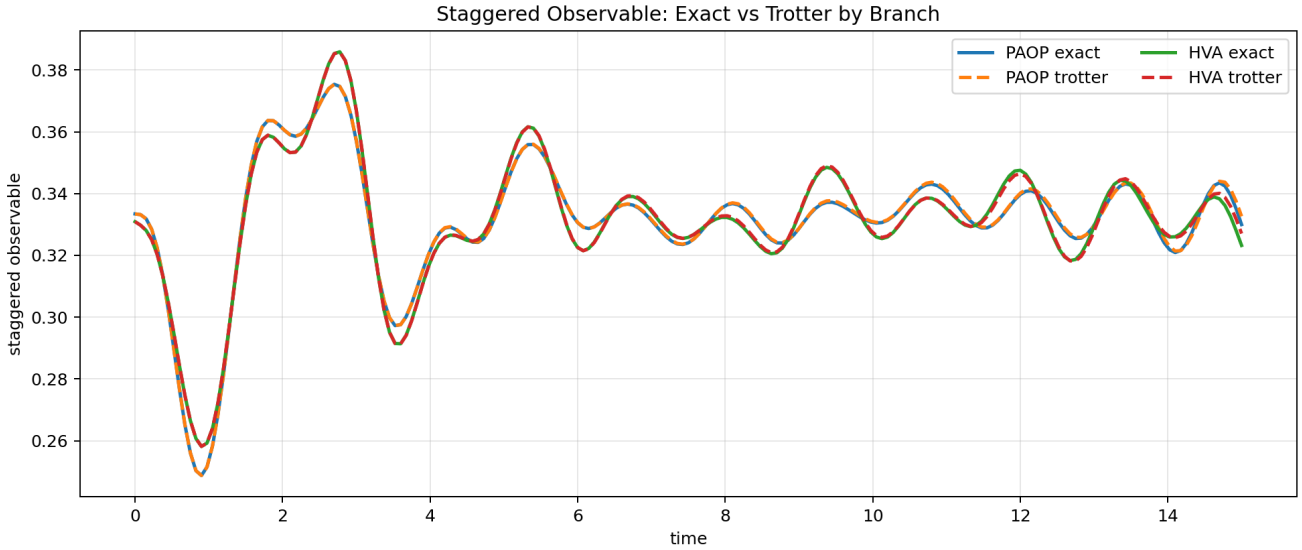

Figure 12: Staggered observable overlay

1.9 9. Interpretable Time-Snapshot Appendix (No Raw Dump)

Snapshot rows every 10th time index for readability (21 rows total).

idx	time	F_paop	F_hva	dE_total_paop	dE_total_hva	dD_paop	dD_hva
0	0.000	0.995909	0.827522	0.000e+00	0.000e+00	0.000e+00	0.000e+00
10	0.750	0.995909	0.827521	1.663e-06	1.320e-06	9.340e-06	1.044e-05
20	1.500	0.995909	0.827517	7.830e-06	1.727e-05	7.975e-06	9.208e-06
30	2.250	0.995904	0.827514	6.163e-06	3.898e-05	5.051e-05	5.763e-05
40	3.000	0.995906	0.827580	1.832e-05	1.211e-04	7.286e-05	8.990e-05
50	3.750	0.995910	0.827601	4.004e-05	4.533e-05	1.343e-04	8.513e-05
60	4.500	0.995910	0.827710	6.052e-05	2.118e-04	2.869e-04	2.994e-04
70	5.250	0.995936	0.827729	1.259e-05	8.408e-05	3.732e-04	3.713e-04
80	6.000	0.995924	0.827590	4.948e-05	2.581e-04	4.923e-04	4.956e-04
90	6.750	0.995951	0.827563	1.030e-04	1.305e-04	3.830e-04	4.241e-04
100	7.500	0.995918	0.827248	1.660e-04	6.827e-04	3.472e-04	3.184e-04
110	8.250	0.995908	0.827536	9.220e-05	8.233e-05	4.452e-04	1.238e-04
120	9.000	0.995863	0.827396	2.033e-04	3.001e-04	7.400e-04	9.808e-04
130	9.750	0.995822	0.827926	1.248e-04	5.640e-04	9.674e-04	1.164e-03
140	10.500	0.995833	0.827907	9.978e-05	1.265e-04	1.854e-03	2.196e-03
150	11.250	0.995769	0.828147	3.302e-04	1.730e-04	6.147e-04	9.901e-04
160	12.000	0.995880	0.828300	2.780e-04	9.137e-04	1.374e-03	1.826e-03
170	12.750	0.995851	0.828023	4.194e-05	1.302e-03	1.309e-04	3.459e-04
180	13.500	0.995954	0.828678	7.711e-04	2.766e-03	3.432e-03	1.780e-03
190	14.250	0.995912	0.827734	1.078e-04	1.247e-03	3.170e-03	3.969e-03
200	15.000	0.995881	0.828071	1.340e-04	7.359e-04	5.556e-03	4.733e-03

1.10 10. Quantile Appendix (Trajectory Error Distributions)

quantile	dE_total_paop	dE_total_hva	dD_paop	dD_hva
min	0.000e+00	0.000e+00	0.000e+00	0.000e+00
q25	1.345e-05	4.970e-05	9.817e-05	8.990e-05
median	6.784e-05	2.426e-04	4.109e-04	3.386e-04
q75	1.702e-04	6.503e-04	1.030e-03	9.851e-04
q90	3.296e-04	1.141e-03	1.963e-03	2.055e-03
q99	9.172e-04	2.538e-03	4.475e-03	4.095e-03
max	1.014e-03	2.766e-03	5.556e-03	4.733e-03

1.11 11. Interpretation Notes

1. Large `nfev` does not guarantee lower `DeltaE`; pool geometry and stopping criteria dominate late-stage behavior.
2. The `42 vs 43` depth discrepancy for B is a provenance difference across two executions (export artifact vs accessibility rerun), not a contradiction.
3. In driven outputs, internal HVA branch quality can differ substantially from imported PAOP branch quality; always interpret by branch source first.
4. This report intentionally removes unreadable raw dumps; all raw values remain in JSON artifacts listed in Section 1.